Detailed Line-by-Line Explanation of DL-1 Jupyter Notebook

This document provides a detailed explanation of the uploaded Deep Learning Jupyter Notebook. Each notebook cell is explained carefully, including: Imports and libraries Dataset handling Model creation Training steps Predictions and evaluation Deep Learning concepts used

Cell 1 (CODE)

Original Content:

```
import pandas as pd
import matplotlib.pyplot as plt
```

Line-by-Line Explanation:

Line	Code	Explanation
1	import pandas as pd	Imports a library/module required for machine learning or data processing.
2	import matplotlib.pyplot as plt	Imports a library/module required for machine learning or data processing.

Cell 2 (CODE)

Original Content:

```
df = pd.read_csv('Boston.csv')
df.head(10)
```

Line-by-Line Explanation:

Line	Code	Explanation
1	df = pd.read_csv('Boston.csv')	Uses Pandas for data handling and manipulation.
2	df.head(10)	General statement or code line.

Cell 3 (CODE)

Original Content:

```
df.drop(columns=['B'],inplace=True)
```

Line-by-Line Explanation:

Line	Code	Explanation
1	df.drop(columns=['B'],inplace=True)	Assigns a value or stores data into a variable.

Cell 4 (CODE)

Original Content:

```
df.isnull().sum()
```

Line-by-Line Explanation:

Line	Code	Explanation
1	df.isnull().sum()	General statement or code line.

Cell 5 (CODE)

Original Content:

```
df.info()
```

Line-by-Line Explanation:

Line	Code	Explanation
1	df.info()	General statement or code line.

Cell 6 (CODE)

Original Content:

```
df.describe()
```

Line-by-Line Explanation:

Line	Code	Explanation
1	df.describe()	General statement or code line.

Cell 7 (CODE)

Original Content:

```
df.corr()['MEDV'].sort_values()
```

Line-by-Line Explanation:

Line	Code	Explanation
1	df.corr()['MEDV'].sort_values()	General statement or code line.

Cell 8 (CODE)

Original Content:

```
X = df.loc[:,['LSTAT','PTRATIO','RM']]
Y = df.loc[:,"MEDV"]
X.shape,Y.shape
```

Line-by-Line Explanation:

Line	Code	Explanation
1	X = df.loc[:,['LSTAT','PTRATIO','RM']]	Assigns a value or stores data into a variable.
2	Y = df.loc[:,"MEDV"]	Assigns a value or stores data into a variable.
3	X.shape,Y.shape	General statement or code line.

Cell 9 (CODE)

Original Content:

```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test = train_test_split(X,Y,test_size=0.25,random_state=10)
```

Line-by-Line Explanation:

Line	Code	Explanation
1	from sklearn.model_selection import train_test_split	Imports a library/module required for machine learning or data processing.
2	x_train,x_test,y_train,y_test = train_test_split(X,Y,test_size=0.25,random_state=10)	Assigns a value or stores data into a variable.

Cell 10 (CODE)

Original Content:

```
from sklearn.preprocessing import StandardScaler
```

Line-by-Line Explanation:

Line	Code	Explanation
1	from sklearn.preprocessing import StandardScaler	Imports a library/module required for machine learning or data processing.

Cell 11 (CODE)

Original Content:

```
scaler = StandardScaler()
```

Line-by-Line Explanation:

Line	Code	Explanation
1	scaler = StandardScaler()	Assigns a value or stores data into a variable.

Cell 12 (CODE)

Original Content:

```
scaler.fit(x_train)
```

Line-by-Line Explanation:

Line	Code	Explanation
1	scaler.fit(x_train)	Trains the neural network using the dataset.

Cell 13 (CODE)

Original Content:

```
x_train = scaler.transform(x_train)
x_test = scaler.transform(x_test)
```

Line-by-Line Explanation:

Line	Code	Explanation
1	x_train = scaler.transform(x_train)	Assigns a value or stores data into a variable.
2	x_test = scaler.transform(x_test)	Assigns a value or stores data into a variable.

Cell 14 (CODE)

Original Content:

```
from keras.models import Sequential
from keras.layers import Dense, Input
```

Line-by-Line Explanation:

Line	Code	Explanation
1	from keras.models import Sequential	Imports a library/module required for machine learning or data processing.
2	from keras.layers import Dense, Input	Imports a library/module required for machine learning or data processing.

Cell 15 (CODE)

Original Content:

```
model = Sequential()
```

Line-by-Line Explanation:

Line	Code	Explanation
1	model = Sequential()	Creates a Sequential deep learning model.

Cell 16 (CODE)

Original Content:

```
model.add(Input(shape=(3,), name='input'))
model.add(Dense(128, activation='relu'))
model.add(Dense(64,activation='relu',name='layer_1'))
model.add(Dense(1,activation='linear',name='output'))
model.compile(optimizer='adam', loss='mse', metrics=['mae'])
model.summary()
```

Line-by-Line Explanation:

Line	Code	Explanation
1	model.add(Input(shape=(3,), name='input'))	Assigns a value or stores data into a variable.
2	model.add(Dense(128, activation='relu'))	Adds a fully connected neural network layer.
3	model.add(Dense(64,activation='relu',name='layer_1'))	Adds a fully connected neural network layer.
4	model.add(Dense(1,activation='linear',name='output'))	Adds a fully connected neural network layer.
5	model.compile(optimizer='adam', loss='mse', metrics=['mae'])	Configures the model with optimizer, loss function, and metrics.
6	model.summary()	General statement or code line.

Cell 17 (CODE)

Original Content:

```
model.fit(x_train,y_train,epochs=100,validation_split=0.05)
```

Line-by-Line Explanation:

Line	Code	Explanation
1	model.fit(x_train,y_train,epochs=100,validation_split=0.05)	Trains the neural network using the dataset.

Cell 18 (CODE)

Original Content:

```
output = model.evaluate(x_test,y_test)
```

Line-by-Line Explanation:

Line	Code	Explanation
1	output = model.evaluate(x_test,y_test)	Evaluates model performance on test data.

Cell 19 (CODE)

Original Content:

```
print(f"Mean Squared Error: {output[0]}"
      ,f"Mean Absolute Error: {output[1]}",sep="\n")
```

Line-by-Line Explanation:

Line	Code	Explanation
1	print(f"Mean Squared Error: {output[0]}"	Displays output to the user.
2	,f"Mean Absolute Error: {output[1]}",sep="\n")	Assigns a value or stores data into a variable.

Cell 20 (CODE)

Original Content:

```
y_pred = model.predict(x=x_test)
```

Line-by-Line Explanation:

Line	Code	Explanation
1	y_pred = model.predict(x=x_test)	Generates predictions using the trained model.

Cell 21 (CODE)

Original Content:

```
print(*zip(y_pred,y_test))
```

Line-by-Line Explanation:

Line	Code	Explanation
1	print(*zip(y_pred,y_test))	Displays output to the user.

Overall Summary:
This notebook demonstrates practical Deep Learning concepts using Python and Jupyter Notebook. It includes: Data preprocessing Neural network model construction Training and optimization Prediction and evaluation Use of deep learning libraries such as TensorFlow/Keras The notebook helps understand how deep learning workflows are implemented step-by-step.